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18/268,823	06/21/2023	Marcus BROOKSHAW	250242-1090	1063

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EXAMINER	
GO, JOHN PHILIP	

ART UNIT	PAPER NUMBER
3681	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/268,823	Applicant(s) BROOKSHAW et al.	
	Examiner JOHN P GO	Art Unit 3681	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 12 January 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-7 and 10-16 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-7 and 10-16 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 12 January 2026.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Status of the Claims

2. **Claims 1-7 and 10-16** are currently pending.

Continued Examination Under 37 CFR 1.114

3. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on January 12, 2026 has been entered.

Information Disclosure Statement

4. The information disclosure statement submitted on January 12, 2026 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement has been considered by Examiner.

Claim Interpretation

5. The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

6. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term "means" or "step" or a term used as a substitute for "means" that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when

the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

7. This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation is: the "gait profiler" recited in **Claims 1-7 and 10-16**.

Because this claim limitation is being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it is being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

If applicant does not intend to have this limitation interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation recites sufficient structure to perform the claimed function so as to avoid it being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 112

8. The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

9. **Claims 1-7 and 10-16** are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

10. Regarding **Claims 1 and 10**, **Claims 1 and 10** recite "in response to the current cumulative risk data exceeding a critical threshold, controlling an actuator of the at least one orthotic device to dynamically adjust a physical characteristic of the at least one orthotic device selected from the group consisting of: bracing force, support angle, and joint restriction." This is considered new matter because the Specification does not disclose any of the aforementioned features. For example, [0063] of the as-filed Specification discloses generating an alarm based on the cumulative risk, but does not disclose comparing the cumulative risk to a threshold, and in response to the risk exceeding the threshold, adjusting the orthotic device, wherein the adjusting includes at least one of bracing force, support angle, and joint restriction.

Additionally, [0047] of the as-filed Specification discloses that an example of the orthotic device may be the device disclosed in PCT/CA2021/051846 ("the '846 application"), but this does not, for example, incorporate the disclosures of the '846 application by reference, and hence cannot provide written description support for the aforementioned claim language. Furthermore, even assuming, *arguendo*, that the disclosures of the '846 application were properly incorporated by reference, the '846 application does not disclose the limitation of adjusting the orthotic device in response to risk data exceeding a critical threshold, wherein the adjusting includes at least one of bracing force, support angle, and joint restriction.

In other words, the present Specification as well as the'846 application disclose, at most, examples of types of orthotic devices, some of which may include actuators. However, neither the present Specification nor the '846 application disclose controlling the actuator to adjust a bracing force, support angle, or joint restriction in response to a cumulative risk exceeding a threshold.

Hence, the feature of adjusting the orthotic device in response to the risk data exceeding a critical threshold as recited in **Claims 1 and 10** constitutes new matter.

11. Dependent **Claims 2-7 and 11-16** are also rejected under 35 U.S.C. 112(a) due to their dependence from **Claims 1 and 10**.

Claim Rejections - 35 USC § 101

12. 35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

13. Claims 1-7 and 10-16 are rejected under 35 U.S.C. 101 because the claimed invention is directed to a judicial exception (i.e., a law of nature, a natural phenomenon, or an abstract idea) without significantly more.

Step 1

Claims 1-7 and 10-16 are within the four statutory categories. Claims 1-7 are drawn to a method for generating a physiological assessment of a user, which is within the four statutory categories (i.e. process). **Claims 10-16** are drawn to a system for generating a physiological assessment of a user, which is within the four statutory categories (i.e. machine).

Prong 1 of Step 2A

Claim 1, which is representative of the inventive concept, recites: ***A method for generating a physiological assessment of a user from biomechanical data gathered from the***

user by a system having one or more processors and at least one orthotic device, the method comprising the steps of:

- ***acquiring biomechanical data of the user to provide mechanical and biomechanical information associated to the user with one or more biomechanical sensors of the at least one orthotic device;***
- ***generating trajectory data of the user using the acquired biomechanical data using a gait profiler and the one or more biomechanical sensors of the at least one orthotic device;***
- ***classifying the trajectory data into variables of interest data according to a set of key biomechanical features, wherein the variables of interest data are processed biomechanical data associated with specific times, postures, and/or activities;***
- ***rejecting variables of interest data that do not fit a pre-determined acceptance criteria, wherein the pre-determined acceptance criteria are cut-off requirements for accepting specific variables of interest as meaningful;***
- ***extracting key features of the non-rejected variables of interest data;***
- ***calculating a current cumulative risk data based on the extracted key features, a user profile covariate data and the variables of interest data;***
- ***generating the physiological assessment of the user using the current cumulative risk data; and***
- ***generating an alarm based on selected current cumulative risk data; and***
- ***in response to the current cumulative risk data exceeding a critical threshold, controlling an actuator of the at least one orthotic device to dynamically adjust a physical***

characteristic of the at least one orthotic device selected from the group consisting of: bracing force, support angle, and joint restriction.

The underlined limitations as shown above, given the broadest reasonable interpretation, **cover the abstract ideas of a certain method of organizing human activity** because they recite managing personal behavior or relationships or interactions between people (i.e. social activities, teaching, and/or following rules or instructions – in this case, the steps of acquiring biomechanical data, generating trajectory data, classifying the trajectory data, rejecting variables that do not fit a pre-determined acceptance criteria, extracting key features, calculating a current cumulative risk data, generating the physiological assessment, and generating an alarm based on risk data include following rules or instructions for assessing a patient's physiological state and notifying a user of the patient condition), e.g. see MPEP 2106.04(a)(2). Any limitations not identified above as part of the abstract idea are deemed "additional elements," and will be discussed in further detail below.

Furthermore, the abstract idea for **Claim 10** is identical as the abstract idea for **Claim 1**, because the only difference between **Claims 1 and 10** is that **Claim 1** recites a method, whereas **Claim 10** recites a system and biomechanical sensors utilized to obtain the biomechanical data.

Dependent **Claims 2-7 and 11-16** include other limitations, for example **Claims 2 and 11** recite sorting and filtering the trajectory data, **Claims 3 and 12** recite formatting the risk data for presentation to the user, **Claims 4 and 13** recite adding additional biomechanical data to the assessment, **Claims 5 and 14** recite types of data utilized in obtaining key biomechanical features, **Claim 6** recites obtaining the biomechanical data from biomechanical sensors, and **Claims 7 and 15-16** recite types of biomechanical sensors and the orientations of the sensors, but these only serve to further narrow the abstract idea, and a claim may not preempt abstract

ideas, even if the judicial exception is narrow, e.g. see MPEP 2106.04, and/or do not further narrow the abstract idea and instead only recite additional elements, which will be further addressed below. Hence dependent **Claims 2-7 and 11-16** are nonetheless directed towards fundamentally the same abstract idea as independent **Claims 1 and 10**.

Prong 2 of Step 2A

Claims 1 and 10 are not integrated into a practical application because the additional elements (i.e. the non-underlined limitations above – in this case, the specific types of data processed by the aforementioned limitations, and the structural limitations utilized to obtain the biomechanical data) amount to no more than limitations which:

- amount to mere instructions to apply an exception – for example, the recitation of biomechanical sensors and an orthotic device, which amounts to merely invoking a computer as a *tool* to perform the abstract idea, e.g. see [0036] and [0047] of the present Specification, see MPEP 2106.05(f);
- generally link the abstract idea to a particular technological environment or field of use – for example, the claim language of biomechanical data utilized to generate a physiological assessment for the user, which amounts to limiting the abstract idea to the field of healthcare/patient monitoring, see MPEP 2106.05(h); and/or
- add insignificant extra-solution activity to the abstract idea – for example, the recitation of adjusting the orthotic device, which amounts to an insignificant application, e.g. see MPEP 2106.05(g).

Additionally, dependent **Claims 2-7 and 11-16** include other limitations, but these limitations also amount to no more than mere instructions to apply an exception (e.g. the types of biomechanical sensors recited in dependent **Claims 6-7 and 15-16**), generally linking the abstract idea to a particular technological environment or field of use (e.g. the types of data recited in dependent **Claims 4-5 and 13-14**), adding insignificant extra-solution activity to the abstract idea (e.g. the formatting and presenting of the data recited in dependent **Claims 3 and 12**), and/or do not include any additional elements beyond those already recited in independent **Claims 1 and 10**, and hence also do not integrate the aforementioned abstract idea into a practical application.

Hence **Claims 1-7 and 10-16** do not include additional elements that integrate the judicial exception into a practical application.

Step 2B

Claims 1 and 10 do not include additional elements that are sufficient to amount to "significantly more" than the judicial exception because the additional elements (i.e. the non-underlined limitations above – in this case, the specific types of data processed by the aforementioned limitations, and the structural limitations utilized to obtain the biomechanical data), as stated above, are directed towards no more than limitations that amount to mere instructions to apply the exception, and/or generally link the abstract idea to a particular technological environment or field of use, wherein the additional elements comprise limitations which:

- amount to elements that have been recognized as well-understood, routine, and conventional activity in particular fields, as demonstrated by:

- The present Specification expressly disclosing that the structural additional elements are well-understood, routine, and conventional in nature:
  - [0036] and [0047] of the Specification discloses that the additional elements (i.e. the biomechanical sensors and the orthotic device) comprise a plurality of different types of generic computing systems;
- Relevant court decisions: The following are examples of court decisions demonstrating well-understood, routine and conventional activities, e.g. see MPEP 2106.05(d)(II):
  - Performing repetitive calculations, e.g. see *Parker v. Flook*, and/or *Bancorp Services v. Sun Life* – similarly, the current invention performs basic calculations (i.e. calculating a risk) and does not impose meaningful limits on the scope of the claims;
  - Storing and retrieving information in memory, e.g. see *Versata Dev. Group, Inc. v. SAP Am., Inc.* – similarly, the current invention recites storing (at least temporarily, in order to acquire) biomechanical data, and retrieving the biomechanical data from storage in order to ultimately generate the physiological assessment; and/or
  - Cutting hair after first determining the hair style, e.g. see *In re Brown* – similarly, the additional elements merely recite adjusting the orthotic device at a high level of generality.

Dependent **Claims 2-7 and 11-16** include other limitations, but none of these limitations are deemed significantly more than the abstract idea because the additional elements recited in the aforementioned dependent claims similarly amount to mere instructions to apply the exception (e.g. the types of biomechanical sensors recited in dependent **Claims 6-7 and 15-16**), generally link the abstract idea to a particular technological environment or field of use (e.g. the types of data recited in dependent **Claims 4-5 and 13-14**), electronic recordkeeping (e.g. the formatting and presenting of the assessment to the user recited in dependent **Claims 3 and 12**), and/or the limitations recited by the dependent claims do not recite any additional elements not already recited in independent **Claims 1 and 10**, and hence do not amount to "significantly more" than the abstract idea.

Hence, **Claims 1-7 and 10-16** do not include any additional elements that amount to "significantly more" than the judicial exception.

Thus, taken alone, the additional elements do not amount to significantly more than the abstract idea identified above. Furthermore, looking at the limitations as an ordered combination adds nothing that is not already present when looking at the elements taken individually, and there is no indication that the combination of elements improves the functioning of a computer or improves any other technology, and their collective functions merely provide conventional computer implementation.

Therefore, whether taken individually or as an ordered combination, **Claims 1-7 and 10-16** are nonetheless rejected under 35 U.S.C. 101 as being directed to non-statutory subject matter.

Claim Rejections - 35 USC § 103

14. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

15. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

16. Claims 1-4, 6-7, 10-13, and 15-16 are rejected under 35 U.S.C. 103 as being unpatentable over Chang (US 2018/0177436) in view of Seyr (US 2012/0215323).

17. Regarding **Claim 1**, Chang teaches the following: ***A method for generating a physiological assessment of a user from biomechanical data gathered from the user by a system having one or more processors and at least one orthotic device, the method comprising the steps of:***

- ***acquiring biomechanical data of the user to provide mechanical and biomechanical information associated to the user with one or more biomechanical sensors of the at least one orthotic device*** (The system includes a biomechanical sensing device (i.e. an orthotic device) including sensors that obtain data including gait metrics and activity data for a

patient (i.e. mechanical and biomechanical information of a user), e.g. see Chang [0023]-[0025].);

- ***generating trajectory data of the user using the acquired biomechanical data using a gait profiler and the one or more biomechanical sensors of the at least one orthotic device*** (The system analyzes the patient gait metrics obtained from the biomechanical sensing device and generates various types of data, for example locomotion biomechanical measurements, step length asymmetries, and shuffle detection, and tremors (i.e. any of which may be interpreted as trajectory data) for the user, e.g. see Chang [0095]-[0096] and [0102].);

- ***classifying the trajectory data into variables of interest data according to a set of key biomechanical features, wherein the variables of interest data are processed biomechanical data associated with specific times, postures, and/or activities*** (The system determines (i.e. classifies) which data should be utilized in the generation of the locomotion biomechanical measurements (i.e. the data utilized is interpreted as "variables of interest"), step length asymmetries, and shuffle detection, and tremors (i.e. any of which may be interpreted as trajectory data), based on (i.e. according to) a set of thresholds (i.e. key biomechanical features), e.g. see Chang [0070], [0083], [0096], and [0112]. Furthermore, the data utilized may include data associated with limping, walking, and/or a risk of falling (i.e. data associated with posture and/or activities), e.g. see Chang [0083], [0096], and [0112], wherein the data associated with walking may include a step cadence based on a right and left ground contact time (i.e. specific times), e.g. see Chang [0071]-[0073].);

- ***rejecting variables of interest data that do not fit a pre-determined acceptance criteria, wherein the pre-determined acceptance criteria are cut-off requirements for accepting specific variables of interest as meaningful*** (The system filters (i.e. rejects) data that does not fit

criteria (i.e. not fitting a pre-determined acceptance criteria), e.g. see Chang [0060], [0070], [0083], [0096], and [0112]. That is, the various criteria for the data are used to determine which data are considered significant (i.e. meaningful).*);*

- ***extracting key features of the non-rejected variables of interest data*** (The system identifies indicators of risk (i.e. key features) from the non-filtered data, e.g. see Chang [0047], [0066], and [0106]-[0109].*);*

- ***calculating a current cumulative risk data based on the extracted key features, a user profile covariate data and the variables of interest data*** (The system utilizes the indicators of risk (i.e. the key features), biometric data (i.e. user profile covariate data), and the non-filtered data (i.e. the variables of interest data) to determine a risk of a user action, for example a fall (i.e. cumulative risk data), e.g. see Chang [0066] and [0106]-[0109].*);*

- ***generating the physiological assessment of the user using the current cumulative risk data*** (The system performs a fall risk assessment that may trigger a response, for example providing the user with an alert, e.g. see Chang [0054], [0108], and [0132], Figs. 2 and 7.*); and*

- ***generating an alarm based on selected current cumulative risk data*** (The system performs a fall risk assessment that may trigger a response, wherein the response may include providing the user with an alert, e.g. see Chang [0054], [0108], and [0132], Figs. 2 and 7.).

But Chang does not teach and Seyr teaches the following:

- ***in response to the current cumulative risk data exceeding a critical threshold, controlling an actuator of the at least one orthotic device to dynamically adjust a physical characteristic of the at least one orthotic device selected from the group consisting of: bracing force, support angle, and joint restriction*** (The system includes an artificial orthotic or

prosthetic joint of a lower extremity (i.e. an orthotic device) including at least one actuator and sensors, e.g. see Seyr [0001]. Furthermore, the sensors determine whether a ground reaction force (GRF) exceeds a threshold, e.g. see Seyr [0016], wherein the GRF is indicative of a user position, for example whether the user is standing or sitting, e.g. see Seyr [0029]-[0030]. Additionally, the actuator, based on the comparison of the measured GRF to the threshold, adjusts a resistance (i.e. a bracing force and/or a joint restriction), e.g. see Seyr [0014]-[0017], [0026], and [0029]-[0033].).

Furthermore, before the effective filing date, it would have been obvious to one ordinarily skilled in the art of healthcare to modify Chang to incorporate adjusting the orthotic device based on a threshold as taught by Seyr in order to enable movement for the user to be as natural as possible under changing conditions, e.g. see Seyr [0003]-[0004].

18. Regarding **Claim 2**, the combination of Chang and Seyr teaches the limitations of **Claim 1**, and Chang further teaches the following: ***The method of claim 1, further comprising the step of:***

- ***sorting and labeling the trajectory data into discrete segments*** (The system may sort the data into step-wise window segments, e.g. see Chang [0070].)***; and***
- ***filtering the discrete segments of trajectory data so as to reduce a number of individual frames and remove noise*** (The segmented data may be analyzed to determine which portions satisfy various conditions/thresholds, for example a minimum amplitude, wherein the data that does not satisfy the conditions is filtered, e.g. see Chang [0060], [0070], [0083], [0096], and [0112].)***;***

- ***wherein the step of classifying the trajectory data into variables of interest data according to a set of key biomechanical features is performed on the filtered discrete segments of trajectory data*** (The data that is not filtered is considered to be usable for determining a user condition, for example a state of mobility including a limp, e.g. see Chang [0060], [0070], [0083], [0096], and [0112].).

19. Regarding **Claim 3**, the combination of Chang and Seyr teaches the limitations of **Claim 1**, and Chang further teaches the following:

- ***The method of claim 1, further comprising the step of formatting and simplifying the risk data of the assessment of the user for presentation to the user*** (The assessed risk for a user action (i.e. the assessment of the user based on the risk data) may be utilized to construct an alert that is displayed to the user, e.g. see Chang [0054], [0108], and [0132], Figs. 2 and 7.).

20. Regarding **Claim 4**, the combination of Chang and Seyr teaches the limitations of **Claim 3**, and Chang further teaches the following:

- ***The method of claim 3, further comprising the step of adding biomechanically derived information to the assessment of the user*** (The data used in assessing the risk of a user action (i.e. the assessment of the user) includes data that is determined (i.e. derived) from raw sensor data, e.g. see Chang [0035], [0052], [0054], [0108], and [0132], Figs. 2 and 7.).

21. Regarding **Claim 6**, the combination of Chang and Seyr teaches the limitations of **Claim 1**, and Chang further teaches the following:

- ***The method of claim 1, wherein the biomechanical data of the user is acquired from biomechanical sensors positioned on the user*** (The data is captured from biomechanical sensing devices positioned on the body of a user, e.g. see Chang [0027]-[0030].).

22. Regarding **Claim 7**, the combination of Chang and Seyr teaches the limitations of **Claim 6**, and Chang further teaches the following:

- ***The method of claim 6, wherein the biomechanical sensors include inertial and angular sensors positioned on a lower-body orthotic device worn by the user*** (The biomechanical sensing devices may be positioned on a user's lower leg, and may include an inertial measurement unit and functionality that measures angular orientation, e.g. see Chang [0027]-[0031].).

23. Regarding **Claim 10**, the limitations of **Claim 10** are substantially similar to those claimed in **Claim 1**, with the sole difference being that **Claim 1** recites a method whereas **Claim 10** recites a system including biomechanical sensors and a processor and memory in communication with the biomechanical sensors. Specifically pertaining to **Claim 10**, Examiner notes that Chang teaches a plurality of biomechanical sensing devices in communication with a processor and a memory, e.g. see Chang [0028], and hence the grounds of rejection provided above for **Claim 1** are similarly applied to **Claim 10**.

24. Regarding **Claims 11-13 and 15-16**, the limitations of **Claims 11-13 and 15-16** are substantially similar to those claimed in **Claims 2-4 and 6-7**, with the sole difference being that **Claims 2-4 and 6-7** recite a method whereas **Claims 11-13 and 15-16** recite a system including

biomechanical sensors and a processor and memory in communication with the biomechanical sensors. Specifically pertaining to **Claims 11-13 and 15-16**, Examiner notes that Chang teaches a plurality of biomechanical sensing devices in communication with a processor and a memory, e.g. see Chang [0028], and hence the grounds of rejection provided above for **Claims 2-4 and 6-7** are similarly applied to **Claims 11-13 and 15-16**.

25. Claims 5 and 14 are rejected under 35 U.S.C. 103 as being unpatentable over the combination of Chang and Seyr in view of Harper (US 2016/0333411).

26. Regarding **Claim 5**, the combination of Chang and Seyr teaches the limitations of **Claim 1**, but does not teach and Harper teaches the following: ***The method of claim 1,***

- ***wherein the key biomechanical features are obtained by forming an indexed combination of curated data from a reference database and an outcome database*** (The system determines criteria for significant and scientific validity for data utilizing various sources, for example SNPedia, NCBI Variation Database (dsSNP), and peer reviewed literature, e.g. see Harper [0207]-[0219] – that is, the data is determined to be significant data (i.e. a key feature) based on analysis of the sources.)***:***

- ***the reference database containing key biomechanical measurements and associated physiological determinant factors, and being constructed using peer-reviewed academic publications*** (The sources include peer reviewed literature including data regarding factors/words and phrases that reveals information that can affect or influence the ability to achieve an ideal weight, e.g. see Harper [0208]-[0209].)***; and***

- ***the outcome database containing linked key biomechanical measurements and associated statistically established outcomes, and being constructed using user monitoring and***

experimentation (The sources include factors (i.e. key measurements) and results (i.e. outcomes) published in peer reviewed journals, e.g. see Harper [0208]-[0209].).

Furthermore, before the effective filing date, it would have been obvious to one ordinarily skilled in the art of patient monitoring to modify the combination of Chang and Seyr to incorporate utilizing the results data from various sources including peer reviewed literature to determine significant data as taught by Harper in order to support the determination made by the system with reliable data, e.g. see Harper [0219].

27. Regarding **Claim 14**, the limitations of **Claim 14** are substantially similar to those claimed in **Claim 5**, with the sole difference being that **Claim 5** recites a method whereas **Claim 14** recites a system including biomechanical sensors and a processor and memory in communication with the biomechanical sensors. Specifically pertaining to **Claim 14**, Examiner notes that Chang teaches a plurality of biomechanical sensing devices in communication with a processor and a memory, e.g. see Chang [0028], and hence the grounds of rejection provided above for **Claim 5** are similarly applied to **Claim 14**.

Response to Arguments

28. Applicant's arguments, see Remarks, filed January 12, 2026, with respect to the newly amended claim limitations of **Claims 1 and 10** have been fully considered but are nonetheless considered new matter under 35 U.S.C. 112(a), for the reasons disclosed above. Applicant must disclose sufficient information to demonstrate that the inventor had possession of the invention at the time of filing and to enable those skilled in the art to make and use the invention, e.g. see MPEP 2162. Applicant acknowledges that the dynamic control of the actuators is required to

implement the disclosed risk assessment, but no language and/or details explaining the dynamic control are disclosed by either the present specification or the '846 application, and hence Applicant has not disclosed sufficient information to demonstrate that the inventor had possession of the invention at the time of filing and to enable those skilled in the art to make and use the invention, and the aforementioned limitations represent new matter. Furthermore, as shown above, in the interest of compact prosecution, the aforementioned limitations have been given full patentable weight but are nonetheless rejected under 35 U.S.C. 103.

29. Applicant's arguments, see Remarks, filed January 12, 2026, with respect to the rejections of **Claims 1-7 and 10-16** under 35 U.S.C. 101 have been fully considered but are not persuasive.

Applicant alleges that the claimed invention is patent eligible because it recites significantly more than an abstract idea, specifically in that it "transforms raw physical inputs into specialized data" that "extends beyond routine or well-understood activities in the art" and "provides a technical improvement over existing data analysis systems…resulting in a more accurate and technically robust output," e.g. see pgs. 9-10 of Remarks – Examiner disagrees.

Examiner notes that "effecting a transformation or reduction of a particular article to a different state or thing" may integrate an abstract idea into a practical application and/or recite significantly more than an abstract idea, e.g. see MPEP 2106.05(c). However, an "article" "includes a physical object or substance," and "changing to a different state or thing usually means more than simply using an article or changing the location of an article," e.g. see MPEP 2106.05(c). The present claims do not claim changing a physical object or substance into a different state or thing, but instead at most claim the manipulation of data, and "for data, mere

"manipulation of basic mathematical constructs [i.e.,] the paradigmatic 'abstract idea,'" has not been deemed a transformation," e.g. see MPEP 2106.05(c). Hence, the claimed invention does not integrate an abstract idea into a practical application and/or does not recite significantly more than an abstract idea on the basis of a particular transformation.

Applicant additionally alleges that the claimed invention is patent eligible because it is not directed towards a certain method of organizing human activities, specifically because it "focuses on the technical transformation of physical biomechanical data into specific physiological metrics," e.g. see pgs. 10-11 of Remarks – Examiner disagrees.

As shown above, the claimed invention covers the abstract idea of a certain method of organizing human activity because the steps of acquiring biomechanical data, generating trajectory data, classifying the trajectory data, rejecting variables that do not fit a pre-determined acceptance criteria, extracting key features, calculating a current cumulative risk data, generating the physiological assessment, and generating an alarm based on risk data include following rules or instructions for assessing a patient's physiological state and notifying a user of the patient condition. Even assuming, *arguendo*, that the aforementioned steps result in the processing of "specific physiological metrics," the absence of *complete preemption* does *not* guarantee that a claim will be eligible, and further notes that preemption is not a stand-alone test for patentability, but rather is inherent in the two-part *Alice/Mayo* framework, e.g. see MPEP 2106.04. That is, claims may not preempt abstract ideas, even if the judicial exception is narrow, e.g. see MPEP 2106.04.

Additionally, [0037] of the as-filed Specification discloses that "existing technology…allow for discrete observations over time, however **they do not have sufficient sensors to track multiple body segments**" (emphasis added). That is, [0037] of the as-filed

Specification discloses that the problem in existing technology is insufficient sensors. The claimed invention does not claim any particular number and/or type of specialized sensor to obtain the data, and hence there is an insufficient nexus between the claimed subject matter and the alleged improvement.

Examiner also notes that the claimed invention is not characterized as reciting a mental process, and hence any arguments pertaining to a mental process are not relevant and/or dispositive to the determination of patent eligibility.

Applicant further alleges that the claimed invention is patent eligible because it is properly analogized to the invention and the resulting rationale in *Desjardins*, e.g. see pgs. 12-13 of Remarks – Examiner disagrees.

As Applicant notes, the invention of *Desjardins* was directed towards improved machine learning training that achieved the improvements of at least reducing storage and preventing catastrophic forgetting. In contrast, the present invention recites obtaining biomechanical data through biomechanical sensors, manipulating the data to obtain a risk, and generating an alarm based on the determined risk. The claimed invention does not recite any particular type of training for a machine learning model, any specific and/or unique type of sensor, and further does not achieve any improvements similar to those disclosed in *Desjardins*.

Applicant also alleges that the claimed invention is patent eligible because [0037] of the as-filed Specification discloses that the claimed invention achieves a technical solution to a problem in biomechanical analysis, and because the claimed invention recites an unconventional arrangement of steps, e.g. se pgs. 13-14 of Remarks – Examiner disagrees.

As stated above, [0037] of the as-filed Specification discloses that the problem in existing technology is insufficient sensors. The claimed invention does not claim any particular number

and/or type of specialized sensor to obtain the data, and hence there is an insufficient nexus between the claimed subject matter and the alleged improvement.

Furthermore, Applicant provides no evidence and/or rationale explaining how the arrangement of limitations is not well-understood, routine, or conventional, beyond a mere conclusory allegation of patentability.

For the aforementioned reasons, **Claims 1-7 and 10-16** are rejected under 35 U.S.C. 101.

30. Applicant's arguments, see Remarks, filed January 12, 2026, with respect to the rejections of **Claims 1-4, 6-7, 10-13, and 15-16** under 35 U.S.C. 102(a)(1) have been fully considered and, in combination with the claim amendments, are persuasive. The rejections of **Claims 1-4, 6-7, 10-13, and 15-16** under 35 U.S.C. 102(a)(1) have been withdrawn. However, for the reasons shown above, **Claims 1-7 and 10-16** are nonetheless rejected under 35 U.S.C. 103.

31. Applicant's arguments, see Remarks, filed January 12, 2026, with respect to the rejections of **Claims 5 and 14** under 35 U.S.C. 103 have been fully considered but are not persuasive and/or are moot.

As an initial matter, Examiners note that **Claims 1 and 10** do not define "an orthotic device" beyond claiming that they include one or more biomechanical sensors. As stated above, Chang teaches a biomechanical sensing device, e.g. see Chang [0023], and hence teaches an orthotic device as presently claimed.

Additionally, as shown above, neither Chang nor Harper is not cited to teach the newly amended limitations pertaining to the actuator of the orthotic device, and instead Seyr is

introduced to teach these limitations. Hence any arguments pertaining to Chang and Harper to teach the newly amended limitations are moot.

For the aforementioned reasons, **Claims 1-7 and 10-16** are rejected under 35 U.S.C. 103.

Conclusion

32. Any inquiry concerning this communication or earlier communications from the examiner should be directed to JOHN P GO whose telephone number is (703)756-1965. The examiner can normally be reached Monday-Friday 9am-6pm Pacific.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, PETER H CHOI can be reached at (469)295-9171. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/JOHN P GO/
Primary Examiner, Art Unit 3681